

Total marks - 120

Attempt Questions 1-8

All questions are of equal value

Answer each question on a NEW PAGE.

Marks

Question 1 (15 marks) Start a NEW page.

(a) Evaluate (i) $\int_0^{\frac{\pi}{4}} \frac{\sec^2 x}{1 + \tan x} \cdot dx$. 2

(ii) $\int_0^{\ln \pi} e^x \sin(e^x) dx$ correct to 2 decimal places. 2

(b) Find (i) $\int \frac{1-x}{\sqrt{1-x^2}} \cdot dx$. 3

(ii) $\int x \cos x \cdot dx$ 2

(c) Find $\int \frac{5t^2 + 3}{t(t^2 + 1)} \cdot dt$. 3

(d) Use the substitution $t = \tan \frac{\theta}{2}$, or otherwise to find

$\int_0^{\frac{\pi}{2}} \frac{2}{1 + \cos \theta} \cdot d\theta$ 3

Question 2 (15 marks) Start a NEW page.

(a)

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- (i) Sketch the graph of $y = \frac{x+3}{x+4}$ showing clearly the coordinates of any point of intersection with the x and the y and the equations of any asymptotes.
- (ii) Use the graph of $y = \frac{x+3}{x+4}$ in part (i) to find the set of values of x for which the function $y = x - \log_e(x+4)$ is increasing.
- (iii) Use the graph of $y = \frac{x+3}{x+4}$ in part (i) to sketch on separate axes

(α) the graph of $|y| = \frac{x+3}{x+4}$.

(β) the graph of $y = \frac{(x+3)^2}{(x+4)(x+3)}$.

- (b) Sketch the curve $y = x^2 + \frac{2}{x}$ showing all essential features.

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Use the graph to determine the nature and number of real roots of the equation $x^3 - kx + 2 = 0$ as ' k ' varies.

Question 3 (15 marks) Start a NEW page.

Marks

- (a) (i) On an Argand diagram shade in the region containing all points representing the complex number z such that

$$|z - (1 + i)| \leq 1 \quad \text{and} \quad |z - (1 + i)| \leq |z|. \quad 2$$

- (ii) Find the exact perimeter of the shaded region. 2

- (b) 6

- (i) Show that the tangent to the rectangular hyperbola $xy = c^2$ at the point $T\left(ct, \frac{c}{t}\right)$ has equation $x + t^2y = 2ct$.

- (ii) The tangents to the rectangular hyperbola $xy = c^2$ at the points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$, where $pq = 1$, intersect at R .

Find the equation of the locus of R and state any restrictions on the values of x for this locus.

- (c) 5

- (i) Show that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

- (ii) Hence show that $\int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx = \int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1 + \tan x}\right) dx$

- (iii) Hence evaluate $\int_0^{\frac{\pi}{4}} \ln(1 + \tan x) dx$

Question 4 (15 marks) Start a NEW page.

- (a) Let $f(t) = t^3 + ct + d$, where c and d are constants.

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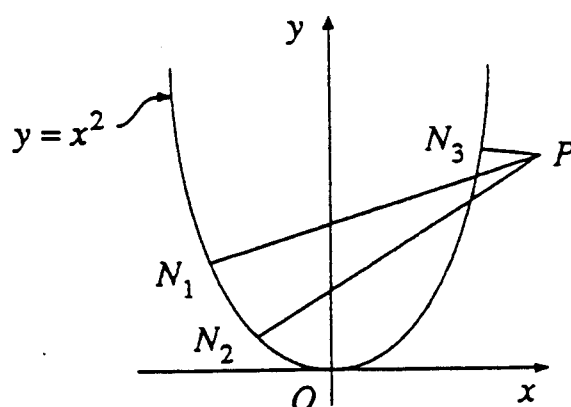
Suppose that the equation $f(t) = 0$ has three distinct real roots, t_1, t_2 and t_3 .

- (i) Find $t_1 + t_2 + t_3$.
- (ii) Show that $t_1^2 + t_2^2 + t_3^2 = -2c$.
- (iii) Since the roots are real and distinct, the graph of $y = f(t)$ has two turning points, at $t = u$ and $t = v$, and $f(u) \cdot f(v) < 0$.

Show that $27d^2 + 4c^3 < 0$.

(b)

5



Consider the parabola $y = x^2$.

Some points (eg P) lie on three distinct normals (PN_1 , PN_2 , and PN_3) to the parabola.

- (i) Show that the equation of the normal to $y = x^2$ at the point (t, t^2) may be written as

$$t^3 + \left(\frac{1-2y}{2}\right)t + \left(\frac{-x}{2}\right) = 0$$

Question 4 continues on page 6

Question 4 (continued)

- (ii) Suppose that the normals to $y = x^2$ at three distinct points $N_1(t_1, t_1^2)$, $N_2(t_2, t_2^2)$, and $N_3(t_3, t_3^2)$ all pass through $P(x_0, y_0)$.

Using the result of part (a) (iii), show that the coordinates of P satisfy

$$y_0 > 3\left(\frac{x_0}{4}\right)^{\frac{2}{3}} + \frac{1}{2}.$$

- (c) For the curve $xy(x + y) + 16 = 0$, show that

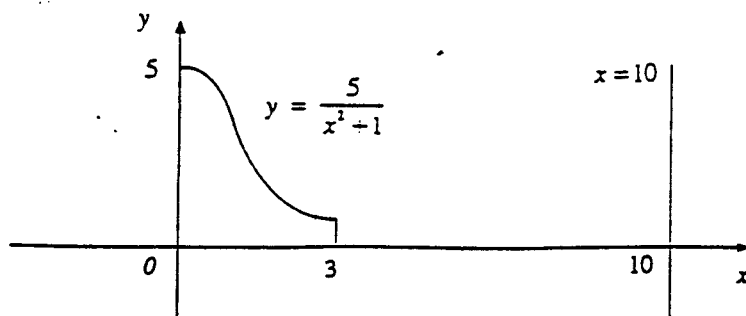
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$$\frac{dy}{dx} = -\frac{(y^2 + 2xy)}{x^2 + 2xy}$$

Hence find the equation of the tangent to the curve $xy(x + y) + 16 = 0$ at the point $(-2, -2)$.

Question 5 (15 marks) Start a NEW page.

(a)



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A circular flange is formed by rotating the region bounded by the curve

$y = \frac{5}{x^2 + 1}$, the x axis and the lines $x = 0$ and $x = 3$, through one complete

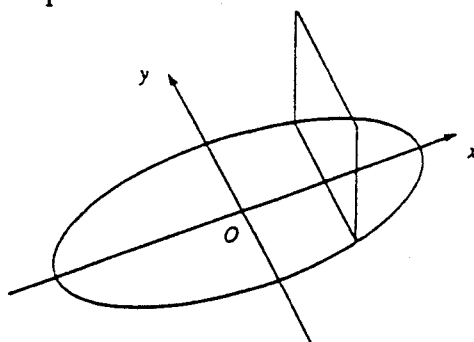
revolution about the line $x = 10$. (All measurements are in centimetres.)

(i) Use the method of cylindrical shells to show that the volume $V \text{ cm}^3$ of the flange is given by $V = \int_0^3 \frac{(100\pi - 10\pi x)}{x^2 + 1} dx$.

(ii) Hence find the volume of the flange correct to the nearest cm^3 .

(b) The base of a tent is in the shape of an ellipse with a major axis of 4 metres and a minor axis of 2 metres. Vertical cross sections taken perpendicular to the major axis of the base are squares.

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(i) If the major axis is taken to lie on the x axis and the minor axis is taken to lie on the y axis, show that the ellipse has equation $\frac{x^2}{4} + y^2 = 1$

(ii) Show that the volume $V \text{ m}^3$ of the tent is given by $V = \int_{-2}^2 (4 - x^2) dx$ and hence find the volume of the tent.

Question 5 continues on page 8

Question 5 (continued)

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(c)

- (i) Find the domain and range of the function $y = \sin(\cos^{-1} x)$.
- (ii) Sketch, showing the important features, the graph of $y = \sin(\cos^{-1} x)$.

Question 6 (15 marks) Start a NEW page.

A body of mass one kilogram is project vertically upwards from the ground at a speed of 20 metres per second. The particle is under the effect of both gravity and a resistance which, at any time, has a magnitude of $\frac{1}{40}v^2$, where v is the magnitude of the particle's velocity at that time.

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In the following questions, take the acceleration due to gravity to be 10 metres per second per second.

- (a) While the body is travelling upwards the equation of motion is

$$\ddot{x} = -\left(10 + \frac{1}{40}v^2\right).$$

- (i) Taking $\ddot{x} = v \frac{dv}{dx}$, calculate the greatest height reached by the particle.
- (ii) Taking $\ddot{x} = \frac{dv}{dt}$, calculate the time taken to reach this greatest height.
- (b) Having reached its greatest height, the particle falls to its starting point. The particle is still under the effect of both gravity and a resistance which, at any time, has a magnitude of $\frac{1}{40}v^2$.
- (i) Write down the equation of motion of the particle as it falls.
- (ii) Find the speed of the particle when it returns to its starting point.

Question 7 (15 marks) Start a NEW page.

(a) Let $I_n = \int_0^{\frac{\pi}{2}} \sin^n x \, dx$ where n is a non-negative integer.

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(i) Show that $I_n = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2} x \cos^2 x \, dx$ when $n \geq 2$.

(ii) Deduce that $I_n = \frac{n-1}{n} I_{n-2}$ when $n \geq 2$.

(iii) Evaluate I_4 .

(b)

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(i) Use de Moivre's theorem and the expansion of $(\cos \theta + i \sin \theta)^3$ to show that $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$.

(ii) Deduce $8x^3 - 6x - 1 = 0$ has solutions $x = \cos \theta$ where $\cos 3\theta = \frac{1}{2}$.

(iii) Find the roots of $8x^3 - 6x - 1 = 0$ in the form $\cos \theta$.

(iv) Hence evaluate $\cos \frac{\pi}{9} \cdot \cos \frac{2\pi}{9} \cdot \cos \frac{4\pi}{9}$.

Question 8 (15 marks) Start a NEW page.

- (a) The acceleration $a \text{ ms}^{-2}$ of a particle P moving in a straight line is given by

6

$$a = 3(1 - x^2),$$

where x metres is the displacement of the particle to the right of the origin.
Initially the particle is at the origin and is moving with a velocity of 4 ms^{-1} .

- (i) Show that the velocity $v \text{ ms}^{-1}$ of the particle is given by

$$v^2 = 16 + 6x - 2x^3.$$

- (ii) Will the particle ever return to the origin? Justify your answer.

- (b) Let n be a positive integer.

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Consider the area bounded by the curve $y = \ln x$, the x axis and $x = n$.
Use integration by parts to show that the value of this area is given by

$$\int_1^n \ln x \, dx = n \ln n - n + 1.$$

Use the trapezoidal rule and ' n ' function values to show that

$$\int_1^n \ln x \, dx \div \frac{1}{2} \ln n + \ln [(n-1)!].$$

Hence deduce that $n! < e\sqrt{n} \cdot \left(\frac{n}{e}\right)^n$.

Question 1

$$\begin{aligned}
 1) \int_0^{\pi} \frac{\sec^2 x}{1+\tan x} dx &= \left[\ln(1+\tan x) \right]_0^{\pi} \\
 &= \ln 2 \\
 2) \int_0^{\ln \pi} e^x \sin e^x dx &= -\left[\cos e^x \right]_0^{\ln \pi} \\
 &= \cos e - \cos e^{\ln \pi} \\
 &= \cos e - \cos \pi \\
 &= \cos e - (-1) \\
 &= 1.54
 \end{aligned}$$

$$\begin{aligned}
 (b) (i) \int \frac{1-x}{\sqrt{1-x^2}} dx &= \int \left(\frac{1}{\sqrt{1-x^2}} - \frac{x}{\sqrt{1-x^2}} \right) dx \\
 &= \sin^{-1} x + \sqrt{1-x^2} + C
 \end{aligned}$$

$$\begin{aligned}
 (ii) \int x \cos x dx &= x \sin x - \int \sin x dx \\
 &= x \sin x + \cos x + C
 \end{aligned}$$

$$\begin{aligned}
 (c) \frac{5x^2+3}{x(x^2+1)} &= \frac{A}{x} + \frac{Bx+C}{x^2+1} \\
 \therefore 5x^2+3 &= A(x^2+1) + Bx+Cx \\
 \therefore 5 &= A+B \\
 3 &= A \Rightarrow B=2 \\
 0 &= C
 \end{aligned}$$

$$\begin{aligned}
 \therefore \int \frac{5x^2+3}{x(x^2+1)} dx &= \int \left(\frac{3}{x} + \frac{2x}{x^2+1} \right) dx \\
 &= 3 \ln x + \ln(x^2+1) + C \\
 &= \ln t^3(t^2+1) + C
 \end{aligned}$$

$$\begin{aligned}
 (d) x &= \tan \theta, \quad dt/d\theta = \sec^2 \theta \\
 \therefore dt &= \frac{2}{1+\tan^2 \theta} d\theta \\
 d\theta &= \frac{2}{1+t^2} dt
 \end{aligned}$$

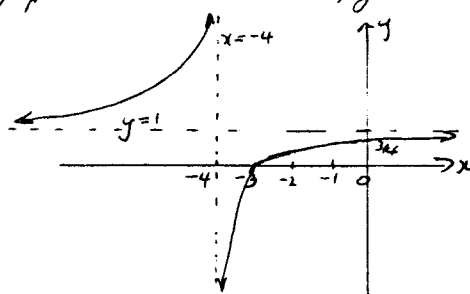
$$\begin{aligned}
 \int_0^{\pi/4} \frac{2}{1+\tan \theta} d\theta &= \int_0^1 \frac{2}{1+t^2} \cdot \frac{2}{1+t^2} dt \\
 &= \int_0^1 \frac{4}{1+t^2+1-t^2} dt \\
 &= \int_0^1 \frac{4}{2} dt \\
 &= \left[2t \right]_0^1 \\
 &= 2
 \end{aligned}$$

$$\begin{aligned}
 \text{OR } \cos \theta &= 2 \cos^2 \theta/2 - 1 \\
 1 + \cos \theta &= 2 \cos^2 \theta/2 \\
 \int_0^{\pi/2} \frac{2 d\theta}{1+\cos \theta} &= \int_0^{\pi/2} \frac{2}{2 \cos^2 \theta/2} d\theta \\
 &= \int_0^{\pi/2} \sec^2 \theta/2 d\theta \\
 &= 2 \left[\tan \theta/2 \right]_0^{\pi/2} \\
 &= 2
 \end{aligned}$$

Question 2

$$\begin{aligned}
 y &= \frac{x+3}{x+4} = \frac{(x+4)-1}{x+4} \\
 &= 1 - \frac{1}{x+4}
 \end{aligned}$$

Asymptotes will be $x=-4, y=1$

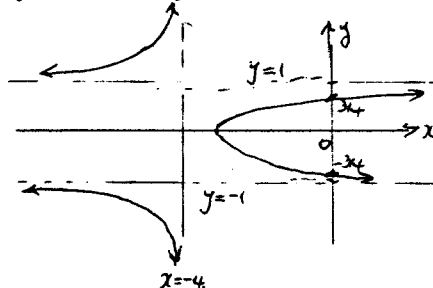


(ii) $y = x - \log_e(x+4)$ is increasing when $dy/dx > 0$

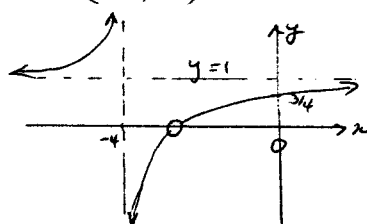
$$\begin{aligned}
 \therefore dy/dx &= 1 - \frac{1}{x+4} \\
 &> 0 \text{ for } x > -3, x < -4
 \end{aligned}$$

but $\log(x+4)$ is defined for $x > -4$ only
 $\therefore$ Curve is increasing for $x > -3$

(iii) if $y \geq 0, |y| = y$ i.e. $y = \frac{x+3}{x+4}$
 $y < 0, |y| = -y$ i.e. $y = -\left(\frac{x+3}{x+4}\right)$



$$(3) y = \frac{(x+3)^2}{(x+4)(x+3)}$$

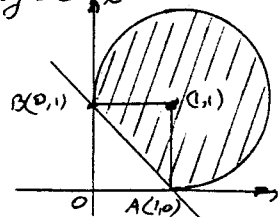


$$\begin{aligned}
 (b) y &= x^2 + \frac{2}{x}, \text{ asymptotes } x=0, y=x^2 \\
 y' &= 2x - \frac{2}{x^2} \\
 &= 0, x^3 = 1 \text{ i.e. at } (1, 3) \\
 y'' &= 2 + \frac{4}{x^3} = 0, x = \sqrt[3]{-2} \therefore \text{P.O.I at } (\sqrt[3]{-2}, 0) \\
 y'''(0) &> 0 \therefore \text{Min. T.P. at } (1, 3)
 \end{aligned}$$

$$\begin{aligned}
 x^3 - kx + 2 &= 0 \\
 x^3 + 2 &= kx \\
 x^2 + \frac{2}{x} &= k \\
 \therefore \text{where } y &= k \text{ intersects } y = x^2 + \frac{2}{x} \text{ is the solution of } x^3 - kx + 2 = 0 \\
 \text{if } k > 3, & \text{ 3 real distinct roots (2 positive, 1 negative)} \\
 k = 3, & \text{ 3 real roots (2 equal roots at } x=1, \text{ 1 negative root at } x=-2) \\
 k < 3, & \text{ 1 real root which is negative.}
 \end{aligned}$$

Question 3

$|z - (1+i)| \leq 1$ is the interior and boundary of the circle centre $(1,1)$ radius 1.
 $|z - (1+i)| = |z|$ is the perpendicular bisector of $(1,1)$ and $(0,0)$ i.e. $x+y=0$
 $\therefore |z - (1+i)| \leq |z|$ is the region on and above $x+y=0$



$$\begin{aligned}
 (i) \text{ PERIMETER} &= AB + \frac{1}{2} \times 2\pi r \\
 &= \sqrt{2} + \frac{1}{2} \times 2\pi \\
 &= \sqrt{2} + \pi
 \end{aligned}$$

$$\begin{aligned}
 (b) y &= \frac{c}{x} \text{ Eqn of tangent} \\
 dy/dx &= -\frac{c}{x^2} \quad y - \frac{c}{x} = -\frac{1}{x^2}(x-ct) \\
 \text{at } x=ct, &= -\frac{1}{x^2} \quad y - ct = -\frac{1}{x^2}(x-ct) \\
 x + \frac{1}{y} &= 2ct
 \end{aligned}$$

$$\begin{aligned}
 (ii) x + \frac{1}{y} &= 2cp \text{ --- tangent at P} \\
 x + \frac{1}{y} &= 2cq \text{ --- tangent at Q} \\
 (p^2 - q^2)y &= 2c(p - q) \\
 y &= \frac{2c}{p+q}
 \end{aligned}$$

$$\begin{aligned}
 x + \frac{2cp}{p+q} &= 2cp \\
 x &= 2cp - \frac{2cp}{p+q} \\
 &= \frac{2cpq}{p+q} \quad (pq=1) \\
 &= \frac{2c}{p+q}
 \end{aligned}$$

$$\therefore R \left(\frac{2c}{p+q}, \frac{2c}{p+q} \right)$$

$\therefore R$ lies on $y=x$
 $\therefore$ locus of R is $y=x$
 where $-c < x < c, x \neq 0$
 if $pq=1$, both tangents are drawn from same point of hyperbola.

$$\begin{aligned}
 (c) \text{ let } u &= a-x \quad x=0, u=a \\
 du &= -dx \quad x=a, u=0 \\
 \int_0^a f(x) dx &= \int_a^0 f(a-u) \cdot (-du) \\
 &= \int_0^a f(a-u) du = \int_0^a f(a-x) dx \\
 \int_0^{\pi/2} \ln(1+\tan x) dx &= \int_0^{\pi/2} \ln(1+\tan(\frac{\pi}{2}-x)) dx \\
 &= \int_0^{\pi/2} \ln\left(1 + \frac{1-\tan x}{1+\tan x}\right) dx \\
 &= \int_0^{\pi/2} \ln \frac{2}{1+\tan x} dx \\
 &= \int_0^{\pi/2} \left[\ln 2 - \ln(1+\tan x) \right] dx \\
 \therefore 2 \int_0^{\pi/2} \ln(1+\tan x) dx &= \left[x \ln 2 \right]_0^{\pi/2} \\
 &= \frac{\pi \ln 2}{2} \\
 \therefore \int_0^{\pi/2} \ln(1+\tan x) dx &= \frac{\pi \ln 2}{4}
 \end{aligned}$$

QUESTION 4 $f(x) = x^3 + cx + d$

$$x_1 + x_2 + x_3 = 0$$

$$\sum x_i x_j = c$$

$$\sum x_i^2 = (\sum x_i)^2 - 2 \sum x_i x_j$$

$$= 0 - 2c$$

$$x_1^2 + x_2^2 + x_3^2 = -2c$$

$$f'(x) = 3x^2 + c$$

$$= 0 \text{ when } x = \pm \sqrt{-\frac{c}{3}}$$

$$x = \pm \sqrt{-\frac{c}{3}}$$

if $f(x) \cdot f(y) < 0$, then $f(\sqrt{-\frac{c}{3}}) f(-\sqrt{-\frac{c}{3}}) < 0$

$$f(\sqrt{-\frac{c}{3}}) = -\frac{c}{3} \sqrt{-\frac{c}{3}} + c \sqrt{-\frac{c}{3}} + d$$

$$= \frac{2c}{3} \sqrt{-\frac{c}{3}} + d$$

$$f(-\sqrt{-\frac{c}{3}}) = \frac{c}{3} \sqrt{-\frac{c}{3}} - c \sqrt{-\frac{c}{3}} + d$$

$$= -\frac{2c}{3} \sqrt{-\frac{c}{3}} + d$$

$$\left(\frac{2c}{3} \sqrt{-\frac{c}{3}} + d \right) \left(-\frac{2c}{3} \sqrt{-\frac{c}{3}} + d \right) < 0$$

$$d^2 - \frac{4c^2}{9} < 0$$

$$i.e. 27d^2 + 4c^3 < 0$$

b) $y = x^2$, $dy/dx = 2x = 2t$ at (t, t^2)

Eqn of normal: $y - t^2 = -\frac{1}{2t}(x - t)$

$$2ty - 2t^3 = -x + t$$

$$x^3 + \left(\frac{1-2t}{2}\right)x + \left(-\frac{x}{2}\right) = 0$$

(x_0, y_0) satisfies eqn of normal so

$$x^3 + \left(\frac{1-2y_0}{2}\right)x + \left(-\frac{x_0}{2}\right) = 0, \text{ this}$$

has 3 distinct roots if

$$27d^2 + 4c^3 < 0 \text{ where } d = -\frac{x_0}{2}$$

$$27\left(-\frac{x_0}{2}\right)^2 + 4\left(\frac{1-2y_0}{2}\right)^3 < 0 \quad c = \frac{1-2y_0}{2}$$

$$(1-2y_0)^3 < -\frac{27x_0^2}{2}$$

$$1-2y_0 < -3 \cdot \frac{x_0^2}{2^{2/3}}$$

$$y_0 > \frac{3x_0^2}{2^{4/3}} + \frac{1}{2}$$

$$y_0 > 3\left(\frac{x_0}{4}\right)^{2/3} + \frac{1}{2}$$

(c) $x^2y + 2xy^2 + 16 = 0$

$$2xy + x^2 \frac{dy}{dx} + y^2 + 2xy \frac{dy}{dx} = 0$$

$$(x^2 + 2xy) \frac{dy}{dx} = -(y^2 + 2xy)$$

$$\therefore \frac{dy}{dx} = -\frac{(y^2 + 2xy)}{x^2 + 2xy}$$

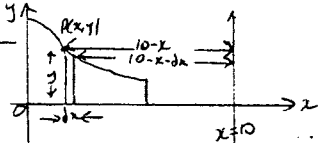
at $(-2, 2)$ $= -\frac{(4+8)}{4+8}$

$$= -1$$

Tangent $y + 2 = -1(x + 2)$

$$i.e. x + y + 4 = 0$$

QUESTION 5



Consider a strip of width dx , x units along the x -axis, when rotated about $x=10$, it generates a thin shelled cylinder of volume dV , where

$$dV = A(x) \cdot dx$$

$$A(x) = \pi(R^2 - r^2), \quad R = 10 - x$$

$$= \pi(R + r)(R - r)$$

$$= \pi(20 - 2x - dx) \cdot dx$$

$$= 2\pi(10 - x)dx, \text{ ignoring } (dx)^2 \approx 0$$

$$\therefore dV = 2\pi(10 - x) \cdot dx$$

$$= 2\pi(10 - x)dx \cdot \frac{5}{x^2 + 1}$$

$$\therefore V = \lim_{dx \rightarrow 0} \sum_{x=0}^3 10\pi \frac{(10 - x)}{x^2 + 1} \cdot dx$$

$$= \int_0^3 \frac{10\pi(10 - x)}{x^2 + 1} \cdot dx$$

$$= \int_0^3 \left(\frac{100\pi}{x^2 + 1} - \frac{10\pi x}{x^2 + 1} \right) dx$$

$$= \left[100\pi \tan^{-1} x - 5\pi \ln(x^2 + 1) \right]_0^3$$

$$= 100\pi \tan^{-1} 3 - 5\pi \ln 10$$

$$\approx 356 \text{ cm}^3$$

(b) General eqn of ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ where } \frac{2a}{2} = 4$$

$$a = 2$$

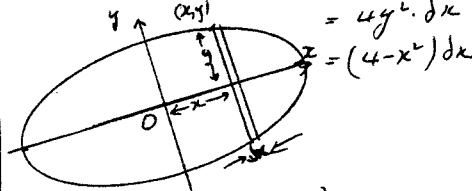
$$i.e. \frac{x^2}{4} + \frac{y^2}{b^2} = 1 \text{ and } \frac{2b}{2} = 2$$

$$b = 1$$

(iii) Consider a slice of width dx , x units along the x -axis and volume dV where $dV = A(x) \cdot dx = (2y)^2 \cdot dx$

$$= 4y^2 \cdot dx$$

$$= (4 - x^2) \cdot dx$$



$$\text{Volume} = \lim_{dx \rightarrow 0} \sum_{x=2}^2 A(x) \cdot dx$$

$$= 2 \int_0^2 (4 - x^2) \cdot dx$$

$$= 2 \left[4x - \frac{x^3}{3} \right]_0^2$$

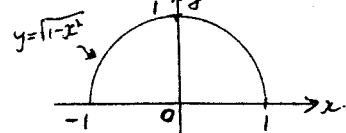
$$= \frac{32}{3} \text{ unit}^3$$

(c) $y = \sin(\cos^{-1} x)$

$$D: -1 \leq x \leq 1 \quad f(x) = \sin(\cos^{-1} x)$$

$$R: 0 \leq y \leq 1 = \sin(\sin^{-1} \sqrt{1-x^2})$$

$$= \sqrt{1-x^2} = \text{Semicircle}$$



$$\text{let } x = \cos^{-1} u$$

$$\cos u = x$$

$$\sin u = \sqrt{1-x^2}$$

$$\therefore x = \cos^{-1} u = \sin^{-1} \sqrt{1-x^2}$$

Question 6

$$F = m\ddot{x}, m = 1$$

$$\ddot{x} = -g - R$$

$$= -(10 + \frac{1}{40}v^2)$$

$$v \cdot \frac{dv}{dx} = -\left(\frac{400 + v^2}{40}\right)$$

$$\frac{dv}{dx} = \frac{400 + v^2}{-40v}$$

$$\therefore dx/dv = \frac{-40v}{400 + v^2}$$

$$\therefore x = \int \frac{-40v}{400 + v^2} \cdot dv$$

$$\text{Max. HEIGHT} = \int_0^{20} \frac{-40v}{400 + v^2} \cdot dv$$

$$= 20 \left[\ln(400 + v^2) \right]_0^{20}$$

$$= 20 \left[\ln 800 - \ln 400 \right]$$

$$= 20 \ln 2 \text{ metres.}$$

(ii) $\frac{dv}{dt} = \frac{400 + v^2}{-40}$

$$\frac{dt}{dv} = \frac{-40}{400 + v^2}$$

$$t = -40 \int \frac{dv}{20^2 + v^2}$$

$$\text{Time} = -40 \int_0^{20} \frac{dv}{20^2 + v^2}$$

$$= 40 \times \frac{1}{20} \left[\tan^{-1} \frac{v}{20} \right]_0^{20}$$

$$= 2 \tan^{-1} 1$$

$$= \frac{\pi}{2} \text{ seconds.}$$

(b) $\ddot{x} = 10 - \frac{1}{40}v^2$

$$v \frac{dv}{dx} = \frac{400 - v^2}{40}$$

$$\frac{dv}{dx} = \frac{400 - v^2}{400 - v^2}$$

$$x = \int \frac{400}{400 - v^2} \cdot dv$$

When particle hits the ground, it has travelled a distance of $20 \ln 2$ and hits ground with velocity v

$$i.e. 20 \ln 2 = \int_0^v \frac{400}{400 - v^2} \cdot dv$$

$$20 \ln 2 = 20 \left[\ln(400 - v^2) \right]_0^v$$

$$\ln 2 = \ln \frac{400}{400 - v^2}$$

$$\therefore 2 = \frac{400}{400 - v^2}$$

$$400 - v^2 = 200$$

$$v^2 = 200$$

$$v = 10\sqrt{2} \text{ ms}^{-1}$$

Question 7

$$\begin{aligned}
 (a) I_n &= \int_0^{\pi} \sin^n x \cdot dx \\
 &= \int_0^{\pi} \sin^{n-1} x \cdot \sin x \cdot dx \\
 &= \left[-\cos x \sin^{n-1} x \right]_0^{\pi} - \int_0^{\pi} \cos x (n-1) \sin^{n-2} x \cdot \cos x \cdot dx \\
 &= \int_0^{\pi} (n-1) \sin^{n-2} x \cdot \cos^2 x \cdot dx \\
 &= (n-1) \int_0^{\pi} (1 - \sin^2 x) \sin^{n-2} x \cdot dx \\
 &= (n-1) \left[\int_0^{\pi} \sin^{n-2} x \cdot dx - \int_0^{\pi} \sin^n x \cdot dx \right] \\
 \therefore I_n &= (n-1) I_{n-2} - (n-1) I_n \\
 n I_n &= (n-1) I_{n-2} \\
 I_n &= \frac{n-1}{n} \cdot I_{n-2} \\
 I_4 &= \frac{3}{4} I_2; I_2 = \frac{1}{2} I_0 \\
 &= \frac{3}{4} \cdot \frac{1}{2} \cdot I_0 \\
 &= \frac{3}{8} I_0; I_0 = \int_0^{\pi} 1 \cdot dx \\
 &= \frac{3\pi}{16}; I_0 = \pi
 \end{aligned}$$

$$\begin{aligned}
 (b) (\cos \theta + i \sin \theta)^3 &= \cos 3\theta + i \sin 3\theta \\
 \therefore \cos 3\theta + i \sin 3\theta &= \cos^3 \theta + 3i \cos^2 \theta \sin \theta + 3 \cos \theta (i \sin \theta)^2 + (i \sin \theta)^3 \\
 \text{Equating reals} \\
 \cos 3\theta &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta \\
 &= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta) \\
 &= 4 \cos^3 \theta - 3 \cos \theta
 \end{aligned}$$

$$(1) 8x^3 - 6x - 1 = 0$$

$$\text{let } x = \cos \theta$$

$$8 \cos^3 \theta - 6 \cos \theta - 1 = 0$$

$$4 \cos^3 \theta - 3 \cos \theta = \frac{1}{2}$$

$$\therefore \cos 3\theta = \frac{1}{2}$$

$$3\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$$

$$\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$$

$$\therefore \text{Roots are } x = \cos \frac{\pi}{9}, \cos \frac{5\pi}{9}, \cos \frac{7\pi}{9}$$

Product of roots are

$$\begin{aligned}
 \cos \frac{\pi}{9} \cos \frac{5\pi}{9} \cos \frac{7\pi}{9} &= \frac{-(-1)}{8} \\
 &= \frac{1}{8}
 \end{aligned}$$

$$\cos \frac{5\pi}{9} = \cos(\pi - \frac{4\pi}{9}) = -\cos \frac{4\pi}{9}$$

$$\cos \frac{7\pi}{9} = \cos(\pi - \frac{2\pi}{9}) = -\cos \frac{2\pi}{9}$$

$$\therefore \cos \frac{\pi}{9} \cdot -\cos \frac{2\pi}{9} \cdot -\cos \frac{4\pi}{9} = \frac{1}{8}$$

$$\therefore \cos \frac{\pi}{9} \cdot \cos \frac{2\pi}{9} \cdot \cos \frac{4\pi}{9} = \frac{1}{8}$$

Question 8

$$(a) a = 3(1-x^2)$$

$$\frac{d}{dx}(\frac{1}{2}v^2) = 3 - 3x^2$$

$$\frac{1}{2}v^2 = \int (3 - 3x^2) dx$$

$$\frac{1}{2}v^2 = 3x - x^3 + C$$

$$x=0 \quad v=4 \quad \begin{cases} 8 = C \end{cases}$$

$$\therefore \frac{1}{2}v^2 = 3x - x^3 + 8$$

$$v^2 = 16 + 6x - 2x^3$$

(11) P starts from $x=0$ and moves to the right, so will it come to rest and change direction?

$$\text{let } P(x) = 16 + 6x - 2x^3$$

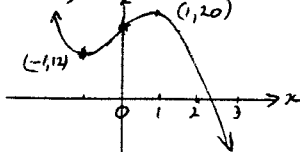
$$P'(x) = 6 - 6x^2$$

$$= 0, \text{ when } x = \pm 1$$

$$P''(x) = -12x$$

$$\therefore \text{Min T.P. at } (-1, 12)$$

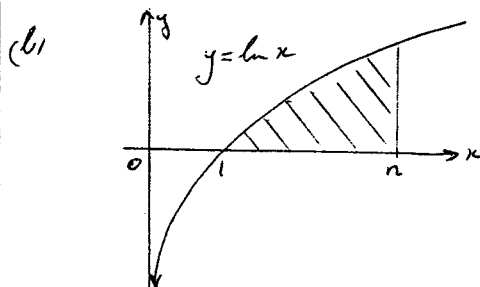
$$\text{Max T.P. at } (1, 20)$$



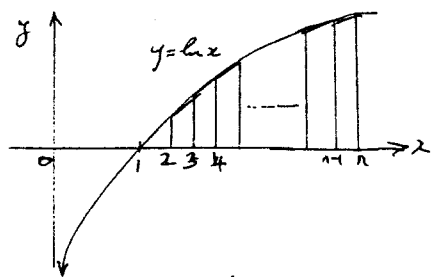
$P(x)$ has only 1 real root

$\therefore P$ comes to rest only once at $2 < x < 3$

so the particle will move to the right where it comes to rest $2 < x < 3$ it then moves to the left without ever stopping and passing through 0 with velocity of -4 m s^{-1} .



$$\begin{aligned}
 \int_1^n \ln x \cdot dx &= x \ln x - \int x \cdot \frac{1}{x} \cdot dx \\
 &= [x \ln x - x]_1^n \\
 &= n \ln n - n - (-1) \\
 &= n \ln n - n + 1
 \end{aligned}$$



by Trapezoidal rule

$$\begin{aligned}
 \int_1^n \ln x \cdot dx &\approx \frac{1}{2} [\text{first} + \text{last} + 2 \text{ others}] \\
 &= \frac{1}{2} [\ln 1 + \ln n + 2(\ln 2 + \ln 3 + \dots + \ln(n-1))] \\
 &= \frac{1}{2} [\ln n + 2 \ln(2 \cdot 3 \cdot 4 \cdot \dots \cdot (n-1))] \\
 &= \frac{1}{2} \ln n + \ln[(n-1)!]
 \end{aligned}$$

as the curve is concave down

Area by Trapezoidal rule < Exact area

$$\therefore \frac{1}{2} \ln n + \ln(n-1)! < n \ln n - n + 1$$

add $\frac{1}{2} \ln n$ to both sides

$$\therefore \ln n + \ln(n-1)! < n \ln n + \frac{1}{2} \ln n - n + 1$$

$$\ln n! < \ln \sqrt{n} \cdot n^n - n + 1$$

$$< \ln \sqrt{n} \cdot n^n - \ln e + \ln e$$

$$< \ln(\sqrt{n} \cdot n^n \cdot e)$$

$$\therefore n! < \sqrt{n} \cdot \left(\frac{n}{e}\right)^n \cdot e$$